Topic: Building systems

Question: Rewrite the system of differential equations in matrix form.

$$x_1' = x_2 - 3x_3$$

$$x_2' = -2x_1 + x_3$$

$$x_3' = 5x_2 - x_1 + 4x_3$$

Answer choices:

A $\vec{x}' = \begin{bmatrix} 0 & 1 & -3 \\ -2 & 0 & 1 \\ -1 & 5 & 4 \end{bmatrix} \vec{x}$

B $\vec{x}' = \begin{bmatrix} 0 & 1 & -3 \\ 2 & 0 & 1 \\ 5 & -1 & 4 \end{bmatrix} \vec{x}$

C $\vec{x}' = \begin{bmatrix} 0 & 1 & 3 \\ 2 & 0 & 1 \\ 1 & 5 & 4 \end{bmatrix} \vec{x}$

D $\vec{x}' = \begin{bmatrix} 0 & 2 & -1 \\ 1 & 0 & 5 \\ -3 & 1 & 4 \end{bmatrix} \vec{x}$

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Solution: A

To write the system in matrix form $\vec{x}' = A\vec{x}$, we define the vector $\vec{x}'$, the matrix A, and the vector $\vec{x}$ as

$$\vec{x}' = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} \qquad A = \begin{bmatrix} 0 & 1 & -3 \\ -2 & 0 & 1 \\ -1 & 5 & 4 \end{bmatrix} \qquad \vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Then we can write the system in matrix form.

$$\vec{x}' = A\vec{x}$$

$$\vec{x}' = \begin{bmatrix} 0 & 1 & -3 \\ -2 & 0 & 1 \\ -1 & 5 & 4 \end{bmatrix} \vec{x}$$

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Topic: Building systems

Question: Convert the fourth order linear differential equation into a system of differential equations.

$$y^{(4)} - y''' + y'' - y' + 3e^t = 0$$

Answer choices:

A $$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 3e^t \end{bmatrix}$$

B $$\vec{x}' = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & -1 & 1 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 3e^t \end{bmatrix}$$

C $$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & -1 & 1 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ -3e^t \end{bmatrix}$$

D $$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ -3e^t \end{bmatrix}$$

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Solution: D

We'll start by defining

$$x_1(t) = y(t)$$

$$x_2(t) = y'(t)$$

$$x_3(t) = y''(t)$$

$$x_4(t) = y'''(t)$$

Now we need to solve the original differential equation for the fourth derivative, $y^{(4)}(t)$.

$$y^{(4)} - y''' + y'' - y' + 3e^t = 0$$

$$y^{(4)} = y''' - y'' + y' - 3e^t$$

Then if we take the derivatives of the equations for $x_1(t)$, $x_2(t)$, $x_3(t)$, and $x_4(t)$, we get

$$x_1'(t) = y'(t) = x_2(t)$$

$$x_2'(t) = y''(t) = x_3(t)$$

$$x_3'(t) = y'''(t) = x_4(t)$$

$$x_4'(t) = y^{(4)}(t) = y''' - y'' + y' - 3e^t = x_4(t) - x_3(t) + x_2(t) - 3e^t$$

Simplifying these equations gives us a system of equations that's equivalent to the original fourth order differential equation.

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$$x_1' = x_2$$

$$x_2' = x_3$$

$$x_3' = x_4$$

$$x_4' = x_2 - x_3 + x_4 - 3e^t$$

And if we wanted to write this nonhomogeneous system as a matrix equation, we'd get

$$\vec{x}' = A\vec{x} + F$$

$$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 1 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ -3e^t \end{bmatrix}$$

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Topic: Building systems

Question: Convert the third order linear differential equation into a system of differential equations in the form $\vec{x}' = A\vec{x} + F$, given $\vec{x}'(0) = B$.

$$y''' - 2y'' + (\cos t)y' - 3y - \sin t = 0$$

$$y(0) = 0,\ y'(0) = 1,\ y''(0) = 3$$

Answer choices:

A $$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 3 & \cos t & 1 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ \sin t \end{bmatrix},\ \vec{x}'(0) = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$$

B $$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 3 & -\cos t & 2 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ \sin t \end{bmatrix},\ \vec{x}'(0) = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

C $$\vec{x}' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & -\cos t & 2 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ \sin t \end{bmatrix},\ \vec{x}'(0) = \begin{bmatrix} 0 \\ -1 \\ -3 \end{bmatrix}$$

D $$\vec{x}' = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & \cos t & 1 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ -\sin t \end{bmatrix},\ \vec{x}'(0) = \begin{bmatrix} -3 \\ 0 \\ -1 \end{bmatrix}$$

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Solution: B

We'll start by defining

$$x_1(t) = y(t)$$

$$x_2(t) = y'(t)$$

$$x_3(t) = y''(t)$$

Now we need to solve the original differential equation for $y'''(t)$.

$$y''' - 2y'' + (\cos t)y' - 3y - \sin t = 0$$

$$y''' = 2y'' - (\cos t)y' + 3y + \sin t$$

Then if we take the derivatives of the equations for $x_1(t)$, $x_2(t)$, and $x_3(t)$, we get

$$x_1'(t) = y'(t) = x_2(t)$$

$$x_2'(t) = y''(t) = x_3(t)$$

$$x_3'(t) = y'''(t) = 2y'' - (\cos t)y' + 3y + \sin t = 2x_3(t) - (\cos t)x_2(t) + 3x_1(t) + \sin t$$

Simplifying these equations gives us a system of equations that's equivalent to the original third order differential equation.

$$x_1' = x_2$$

$$x_2' = x_3$$

$$x_3' = 3x_1 - (\cos t)x_2 + 2x_3 + \sin t$$

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Rewrite the initial conditions.

$$y(0) = 0 \qquad x_1(0) = 0 \qquad x_1'(0) = 1$$

$$y'(0) = 1 \qquad x_2(0) = 1 \qquad x_2'(0) = 3$$

$$y''(0) = 3 \qquad x_3(0) = 3 \qquad x_3'(0) = 5$$

And if we wanted to write this nonhomogeneous system as a matrix equation, we'd get

$$\vec{x}' = A\vec{x} + F \qquad \vec{x}'(0) = B$$

$$\vec{x}' = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 3 & -\cos t & 2 \end{bmatrix} \vec{x} + \begin{bmatrix} 0 \\ 0 \\ \sin t \end{bmatrix} \qquad \vec{x}'(0) = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

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